



EN

Operating Instructions Vacuum-Operation-Center VOC-AD-S-63/100

Please note

These operating instructions were prepared in German.
Store for future use.

Subject to technical changes, typographical and other errors.

Publisher

© J. Schmalz GmbH, 08.2018

This document is protected by copyright law. All resulting rights remain the property of J. Schmalz GmbH. The reproduction of this document or parts thereof is only permitted within the legal restrictions established by copyright law. Any amendment or abridgement of the document is prohibited without the express written consent of J. Schmalz GmbH.

Contact

J. Schmalz GmbH
Johannes-Schmalz-Str. 1
D-72293 Glatten

Tel. +49 (0)7443 2403-0
Fax +49 (0)7443 2403-259
schmalz@schmalz.de
www.schmalz.com

You can find contact information on the Schmalz companies and trading partners worldwide at
 www.schmalz.com/vertriebsnetz

1	Safety Information	5
1.1	Classification of the safety information	5
1.2	Warnings	6
1.3	Mandatory action signs	6
1.4	General safety information	7
1.5	Intended use	9
1.6	Requirements and information relating to the installation, maintenance and operating personnel	10
2	Product Description	12
2.1	Components of the Vacuum-Operation-Center VOC-AD-S-63/100	12
2.2	General description of the functions	13
2.2.1	Standard machining	14
2.2.2	Draining the storage tank during machining operation	15
2.2.3	Falling below the vacuum limit	15
2.2.4	Critical fill level reached	16
2.2.5	EMERGENCY STOP	16
3	Technical Data	17
3.1	Vacuum-Operation-Center VOC-AD-S-63/100	17
3.2	Pneumatic Connection Diagram	18
4	Delivery, Packaging and Transport	19
4.1	Delivery	19
4.1.1	Scope of delivery	19
4.1.2	Check integrity	19
4.1.3	Report damage	19
4.2	Packaging	19
4.2.1	Inland (land transport)	20
4.2.2	Abroad (overseas or air freight)	20
4.3	Transport	20
4.3.1	Shipment	20
4.3.2	Assembly and removal of the transport locks	20
5	Commissioning	22
5.1	Installation	22
5.1.1	Electrical connection	23
5.1.2	Connecting the signal line to the overall machine	24
5.1.3	Connecting the vacuum and the liquid feedback	25
5.1.4	Connecting the compressed air supply	25
5.2	Test run	26
6	Operation	27
7	Troubleshooting	28
8	Maintenance	30

8.1	General safety information	30
8.2	Pump oil maintenance	31
8.3	Service table	32
8.4	Cleaning and replacing the fine filter	34
9	Spare and Wear Parts	35
9.1	Warranty	35
9.2	Spare and Wear Parts for VOC-AD-S-63/100	36
10	Accompanying Documents	39
10.1	Vacuum Switch Operating Instructions	39
10.2	Vacuum Generator Operating Instructions	39
10.3	Declaration of Conformity	39

1 Safety Information

1.1 Classification of the safety information

Danger

This warning warns of danger that will result in death or serious injuries if it is not avoided.

 DANGER	
	Type and source of the danger Consequences of the danger ► Removing the danger

Warning

This warning warns of danger that may result in death or serious injuries if it is not avoided.

 WARNING	
	Type and source of the danger Consequences of the danger ► Removing the danger

Caution

This warning warns of danger that may result in injuries if it is not avoided.

 CAUTION	
	Type and source of the danger Consequences of the danger ► Removing the danger

Attention

This warning warns of danger that may result in property damage if it is not avoided.

ATTENTION	
	Type and source of the danger Consequences of the danger ► Removing the danger

General information

This symbol is used if important information on handling the machine/system/unit is provided.

	Note / Information
---	--------------------

1.2 Warnings

Explanation of the warning signs used in these operating instructions.

Icon	Description	Icon	Description
	General warning		Warning of damage to hearing
	Warning of electrical voltage		Warning of flying parts
	Warning of hand injuries		Warning of risk of crushing
	Warning of suspended load		Warning of vacuum
	Warning of hot surface		Warning of environmental damage

1.3 Mandatory action signs

Explanation of the mandatory action signs used in these operating instructions.

Icon	Description	Icon	Description
	Follow the operating instructions		Wear safety boots
	Wear gloves		Remove mains plug
	Wear face mask		Wear eye protection
	Disconnect prior to maintenance or repair		Wear hearing protection
	Use head protection		

1.4 General safety information

 WARNING	
 	Risk of injury as a result of non-compliance with the general safety Information Leads to personal injury and damage to the system ▶ The operating instructions contain important information on handling the Vacuum-Operation-Center. Please read the operating instructions carefully and store them for future use. ▶ The operating instructions are aligned to the equipment delivered by Schmalz. Customer changes to the system are not taken into account and are generally prohibited! ▶ The Vacuum-Operation-Center may only be connected and commissioned once the operating instructions have been read and understood! ▶ Unauthorised conversions and changes to the Vacuum-Operation-Center are prohibited for reasons of safety! ▶ In general, the components must be protected against damage of any kind! ▶ The operating instructions must be stored with the machine. It must be ensured that all persons performing activities on the machine are able to consult the operating instructions at all times. ▶ The operating instructions must be provided together with the operating manual prescribed by the Occupational Health and Safety Act and the Work Equipment Regulations (switching function of the machine control system). ▶ In particular, the operator must ensure that additional risks that arise as a result of the special operating conditions at the respective operating site and the integration of the Vacuum Operation Center into a machine pursuant to Machinery Directive 2006/42/EC are identified as part of a risk analysis (in terms of Occupational Health and Safety Act § 5)

 WARNING	
 	Risk of injury from vacuum and high volume flow Hair, skin, body parts and items of clothing will be sucked in. ▶ Wear close-fitting clothing and safety goggles. ▶ Use a hair net. ▶ Do not look or reach into the suction connection. ▶ Ensure that the suction opening is kept away from bodily orifices.

 CAUTION	
 	Noise pollution when disconnecting the compressed supply Damage to hearing ▶ Wear hearing protection
 CAUTION	
	Risk of burns from contact with hot surfaces Burns to the skin ▶ Do not touch hot surfaces ▶ Only perform maintenance work after allowing for a 1h cooling period
 CAUTION	
	Risk of eye injuries All vacuum generators generate an exhaust air flow. Depending on the purity of the ambient air, this may contain exhaust gas particles that pass out of the exhaust opening at high speed and cause face and eye injuries. ▶ Keep away from the exhaust airflow. ▶ Wear safety goggles.
 CAUTION	
 	Risk of eye and airway injuries The vacuum pump exhaust gases may form aerosols (air-particle mixtures) that may result in eye and airway injuries. ▶ Wear a face mask. ▶ Wear safety goggles.
 CAUTION	
	Risks of using the vacuum clamping system as part of an integrated system at the operating site Property damage and/or personal injury ▶ Prepare a risks analysis for the entire system with regard to the vacuum clamping system.

1.5 Intended use

The Vacuum-Operation-Center (VOC) is intended to control and maintain a specific vacuum and act as a liquid separator.

Appropriate safety precautions on systems and units that are used in connection with the VOC must prevent any risks to individuals, animals or property as a result of a loss of operating vacuum.

A machine or system pursuant to Machinery Directive 2006/42/EC with the relevant safety installations is required in order to use the VOC.

The VOC is exclusively intended to control and maintain a specific operating vacuum that is used to fasten certain workpieces to an appropriate clamping device or to handle appropriate workpieces with the relevant handling system.

The VOC is specifically designed for evacuating media with a liquid content and can be used to generate a vacuum for machinery in wet machining areas. Chips or other small solids contained in the separated liquid can be separated from the evacuated medium and do not impair the function of the VOC in small quantities.

The water content in the liquid may not fall below 70%.

The integration of the VOC into the overall machine is the responsibility of the manufacturer of the overall machine or the machining centre. The VOC is equipped with supply lines, sensors and actuators.

Before commissioning, the VOC supply lines must be connected in accordance with the descriptions provided in Chapter 5 and the transmitting electrical lines must be connected to the overall machine control unit via a multi-pole plug. The manufacturer of the overall machine is responsible for the further processing of these electrical signals.

For example, the use of the VOC signals must ensure that the machine release to process workpieces is interrupted when the supplied liquid reaches the critical fill level and if the vacuum level falls below -600 mbar.

Any other use that deviates from the intended use is not permitted and may lead to dangerous situations!

The VOC is:

- not intended to suck in body parts and other living creatures and
- not designed for mechanical loads of any kind.

The evacuation of items with an implosion risk and unauthorised media is prohibited in any form.

Unauthorised conversions and changes to the VOC are prohibited for reasons of safety!

1.6 Requirements and information relating to the installation, maintenance and operating personnel

The VOC was designed and constructed in consideration of a risk analysis and after careful selection of the harmonised standards to be complied with, as well as additional technical specifications. This represents a state-of-the-art design and ensures maximum safety.

However, this level of safety can only be reached in practice if all the necessary measures have been taken. The company operating the VOC has a duty of care to plan these measures and control their implementation.

The VOC may only be installed and maintained by qualified experts, mechanics and electricians. Work on the electronics may generally only be performed by electricians.

An expert is any individual whose technical training, knowledge and experience as well as their knowledge of the applicable provisions allows them to assess the activities assigned to them, identify possible risks and take appropriate safety measures. An expert must comply with the applicable technical regulations.



Any individual assigned to install, commission, operate, maintain and repair the VOC by the operating company must have read and understood the operating instructions.



DANGER



Electric shock from contacting live parts

Risk of death due to electric shock

- ▶ Work on the electrical installation may only be performed by an electrician.
- ▶ Disconnect the electrical energy supply prior to assembly, adjustment and maintenance activities and secure against restart.
- ▶ Ensure that the power has been disconnected, earthed and short-circuited.
- ▶ Do not connect or disconnect a live mains plug.

The operating company must introduce internal measures to ensure that

1. the VOC is only used as intended.
2. the VOC is only operated in a flawless and functional state and that, in particular, the functionality of the safety installations is reviewed on a regular basis.
3. the operating instructions are always complete and available in a legible condition at the machine's operating location.
4. only qualified and authorised personnel operate, maintain and repair the VOC.
5. the operating personnel is regularly instructed on all issues of occupational health and safety and environmental protection and that they are familiar with the operating instructions and particularly the relevant safety information.
6. the operating vacuum is only activated once the vacuum clamping system is sealed against the workpiece, as incorrect use may impair the vacuum in the system or result in a complete failure.

7. the vacuum in the system may be impaired when using the VOC as the central vacuum generator for several clamping devices if several long hose lines have been installed. In general, the line lengths of the pneumatic and vacuum connections must be kept as short as possible!
8. the VOC is not used together with explosive material or in explosive areas!
9. the VOC is placed on a solid and level floor!

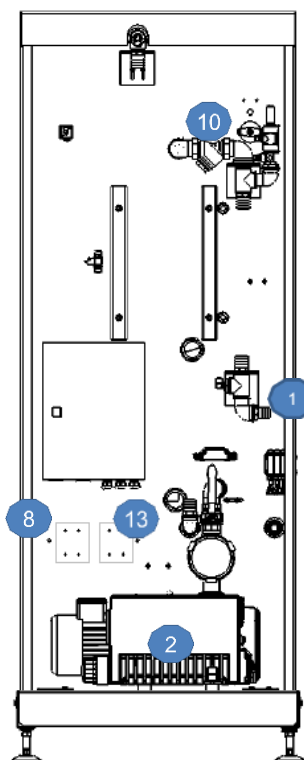
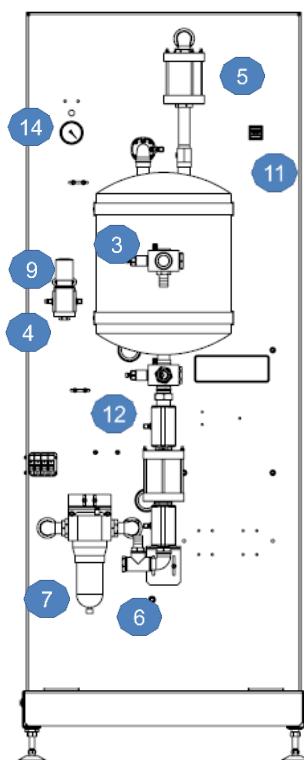
The responsibilities for the various activities on the VOC must be clearly defined and complied with. No ambiguous competencies are permitted.

Trainee operating personnel may initially only work with the VOC under the supervision of an experienced person.

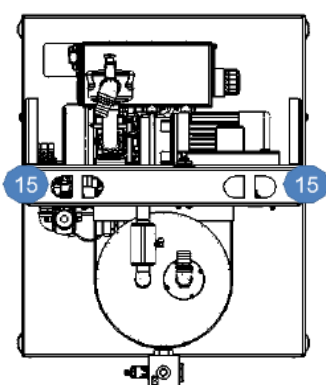
Written confirmation must be provided of any completed and successful training and instruction.

2 Product Description

2.1 Components of the Vacuum-Operation-Center VOC-AD-S-63/100



Item	Description
1	Vacuum connection
2	Vacuum generator/pump
3	Fill level sensor ("Critical fill level") and inspection glass
4	Compressed air connection (4 bar)
5	Fine filter
6	Connection thread G3/4" for hose connection
7	Droplet separator
8	Multi-pole plug X1 power supply
9	Pressure regulator
10	Prefilter
11	Vacuum switch/sensor
12	Fill level sensor "Start feedback"
13	Multi-pole plug X2 signal control
14	Display vacuum level in the tank
15	Transport lugs



2.2 General description of the functions

The VOC is used to control and maintain a specific operating vacuum and as a liquid separator for evacuated liquid media with a low to medium level of pollution.



Please note that the liquid in the storage tank can only drain out if the liquid line ends below the level of item [6]!

The VOC provides an interface between the machine and the clamping system and generates the retention force between the clamping device and workpiece together with the clamping device. The media evacuated as a result of the vacuum generation is fed from the clamping device through the VOC.

Liquids contained in the evacuated medium are separated by the VOC and collected in the storage tank. This stops liquids from entering the vacuum generator, item [2].

The entry of liquids damages the vacuum generator, item [2], and may even result in a complete failure.

The liquid that collects in the VOC as a result of the evacuation of the vacuum clamping device is stored in the bottom glass container. If the fill volume exceeds approx. 1.5 - 2 litres, fill level sensor [2Z1], item [12], (cf. Chap. 3.2) sends a signal to the system that the feedback process can be started.

Fill level sensor [2Z1], item [12], continues to send a signal until the liquid level in the storage tank falls below this range. The liquid feedback process stops once fill level sensor [2Z1], item [12], stops emitting a signal. This means that the storage tank has been sufficiently drained.

Under the following operating conditions the loss of the vacuum results in the loss of the retention force on the workpiece and means that the workpiece is no longer clamped in place.

 WARNING	
	Loss of clamping force due to a loss of vacuum Personal injury and property damage due to workpieces accelerated by machining forces ▶ This must be taken into account when designing the overall machine. ▶ Individuals are not permitted to stay in the hazard area during the machining process.

If more liquid is separated from the evacuated medium than can be conveyed out of the VOC by the feedback process, the storage tank will constantly fill with liquid to the critical level despite continuous draining. This condition is detected and signalled to the machine control unit by fill level sensor [1Z1], item [3], when the container is half full ("Machine/Coolant supply STOP").

In order to prevent the liquid from reaching the vacuum generator, item [2], the storage tank must be drained completely when this signal is emitted. The machining of the workpiece and the supply of liquid in the CNC processing centre must be at a complete stop as long as the signal continues to be emitted by fill level sensor [1Z1], item [3], in order to ensure that no additional liquids are sucked in.

The vacuum switch, item [11], monitors the operating vacuum and releases the machining operation when the pressure difference equals or exceeds -600 mbar. If the level falls below this limit, the machine receives an electronic signal from the vacuum switch: "Machine STOP".

The entire system is ventilated (=> pressure equalisation) as soon as the vacuum pump is switched off and the valves responsible for the feedback process are fully opened. This accelerates the complete discharge and the liquid is conveyed directly out of the storage tank.

 WARNING	
	Loss of clamping force due to a loss of vacuum Personal injury and property damage due to workpieces accelerated by machining forces ▶ This must be taken into account when designing the overall machine. ▶ Individuals are not permitted to stay in the hazard area during the machining process.

2.2.1 Standard machining

During the standard machining process, the VOC is evacuated before the workpiece on the machine table or the clamping device can be clamped by the vacuum. This requires the vacuum generator to be operated. The workpiece is then clamped.

The release for machining the workpiece is provided once the required vacuum value of -600mbar is reached and displayed and signalled by vacuum switch [1Z2], item [11] (The vacuum switch does not send a current signal (0) more.).

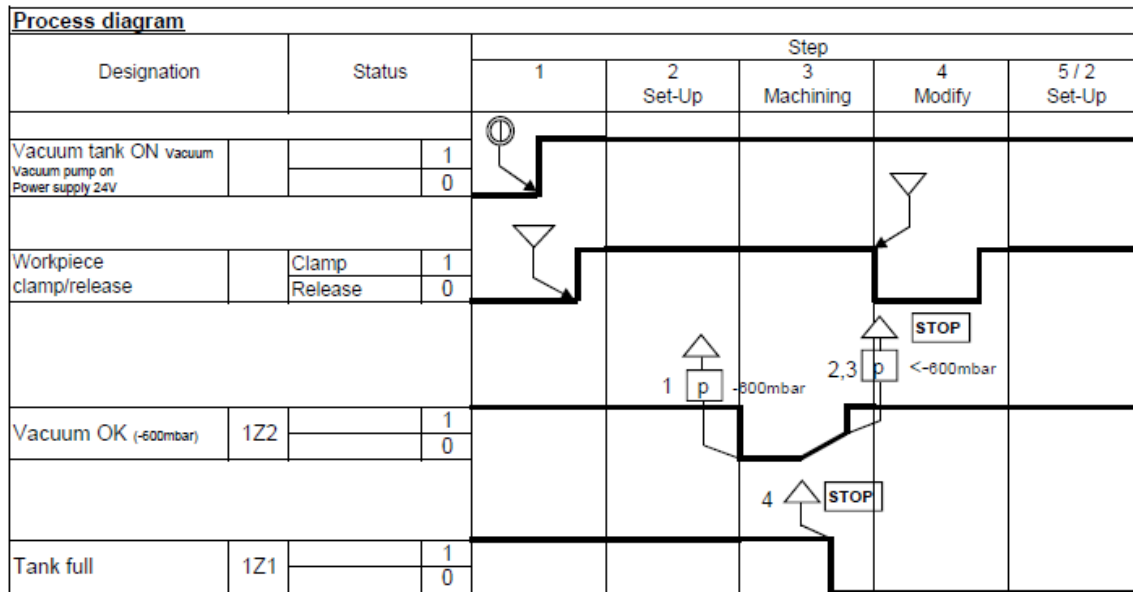
The precise process is displayed in the flow diagram.

Electronic signals emitted by the VOC described in the flow diagram:

Switch Vacuum generator on (step 1)

Clamp / release workpiece (step 2 u. 4: set up)

- △ 1: Release for machining from -600mbar (see "Vacuum OK" designation)
- △ 2: Minimum vacuum level not reached (-600mbar); machining/coolant supply STOP (see "Vacuum OK" designation)
- △ 3: A signal, vacuum < -600mbar, is sent to the machine (see "Vacuum OK" designation)
- △ 4: Maximum permitted liquid level reached; machining/coolant supply STOP (see "Tank full" designation)



2.2.2 Draining the storage tank during machining operation

The liquid, which accumulates by evacuating the vacuum clamping device in VOC, accumulates on the glass storage tank. During the standard machining operation fills the reservoir. From a fill level of about 1.5 - 2 liters, the level sensor [ZZ1] item [12] transmits a signal to the machine control from the VOC (state change of current signal (1) to no current signal (0)).

The recirculation process can be started.

Thereafter, the valves [2V1] and [2V2] are switched alternately.

In order not to lose the operating vacuum during machining, it must be ensured that valve [2V2] is closed before valve [2V1] is opened with a time delay. The liquid is drained from the system by alternately switching the two valves.

The interval time for valve [2V1] amounts to 15 seconds closed/6 seconds open. Valve [2V2] is switched to alternate with valve [2V1]. This function continues until fill level sensor [ZZ1], item [12], stops emitting a signal. This means that the system has been drained and both valves move back to their starting position.

2.2.3 Falling below the vacuum limit

Vacuum switch [1Z2], item [11], emits a signal if the necessary operating vacuum drops below -600mbar during the machining process (Signal state changes from (0) to (1)). The workpiece machining process is immediately stopped from the machine control unit, as the clamping system can no longer provide the necessary clamping force which may cause a dangerous situation.

Valve [1V1] must remain open and valve [1V2] must remain closed ("Clamp workpiece" status) to ensure that the workpiece cannot be released unexpectedly.

The system must now be inspected for leakages.

 WARNING	
	Loss of clamping force due to a loss of vacuum Personal injury and property damage due to workpieces accelerated by machining forces ▶ This must be taken into account when designing the overall machine. ▶ Individuals are not permitted to stay in the hazard area during the machining process.

The workpiece machining process can only be resumed once an operating vacuum of -600mbar is established!

2.2.4 Critical fill level reached

Sensor [1Z1], item [3], emits a signal (Signal state changes from (1) to (0)) if the liquid level rises so dramatically during the machining process that the liquid can no longer be adequately discharged.

This signal has to stop the machining process as well as the supply of coolant. Valve [1V1] must once again be opened and valve [1V2] must remain closed in this case.

The storage tank is completely drained (see Chap. 2.3.2 "Draining the storage tank during machining operation") and the machining process can be resumed as soon as sensor [2Z1], item [12], no longer sends a signal from the VOC to the machine control unit, which means that the tank is completely empty! The machining release and the "Machine stop" signal reset takes place at the same time.

 WARNING	
	Loss of clamping force due to a loss of vacuum Personal injury and property damage due to workpieces accelerated by machining forces ▶ This must be taken into account when designing the overall machine. ▶ Individuals are not permitted to stay in the hazard area during the machining process.

2.2.5 EMERGENCY STOP

If a malfunction occurs during a stage of the center's life (installation, commissioning, operation, maintenance, troubleshooting), individual components fail or a sudden loss of vacuum occurs for another reason, the VOC must be shut down using an EMERGENCY STOP integrated into the overall machine or the CNC machining centre.

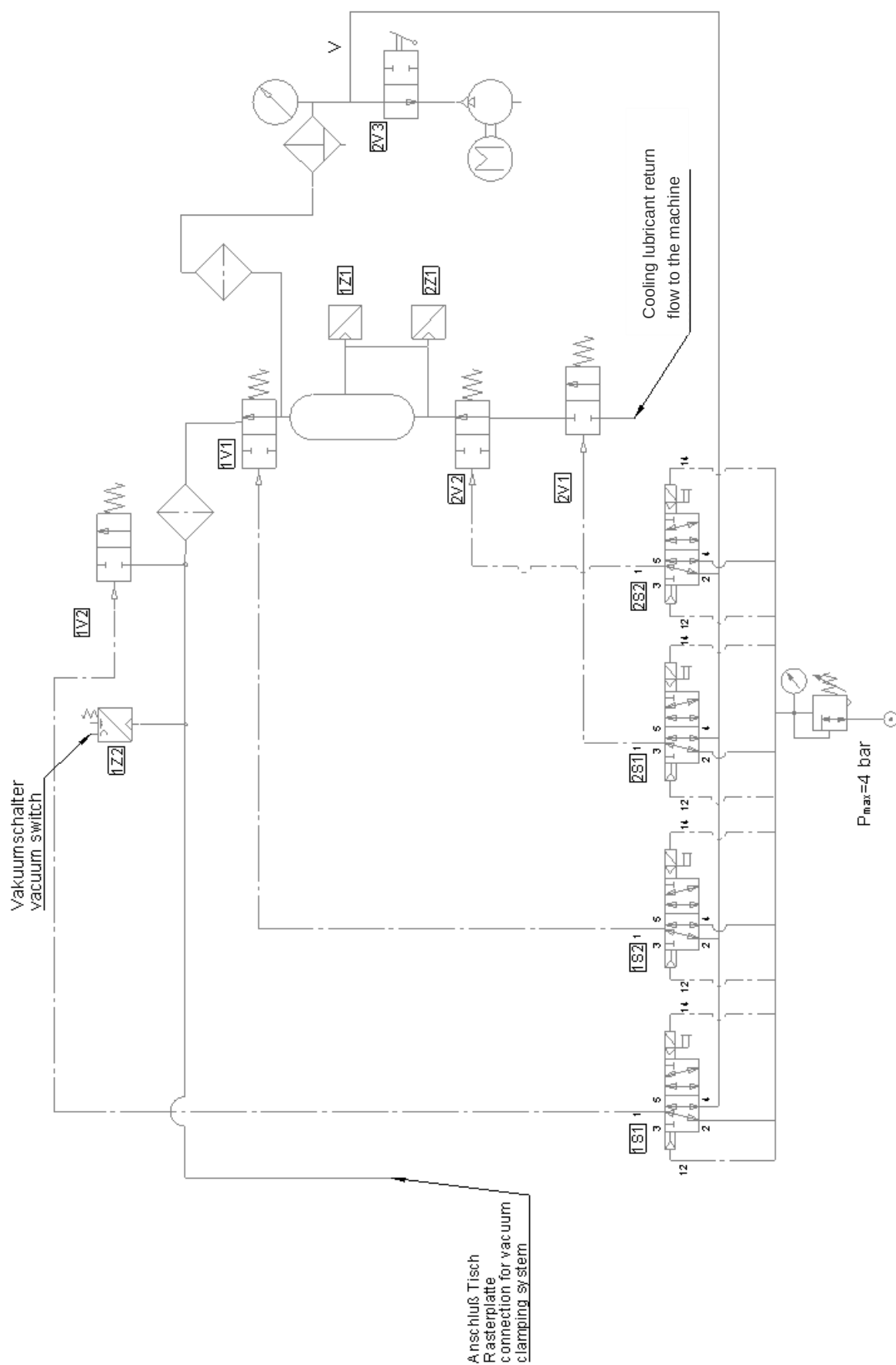
 WARNING	
	Loss of clamping force due to a loss of vacuum Personal injury and property damage due to workpieces accelerated by machining forces
	▶ This must be taken into account when designing the overall machine. ▶ Individuals are not permitted to stay in the hazard area during the machining process.

3 Technical Data

3.1 Vacuum-Operation-Center VOC-AD-S-63/100

Feature	Description
Dimensions [L x W x H]	Approx. 850 x 1100 x 2200 mm
Installation position	Upright, vertical +/- 5°
Vacuum connections	Internal hose diameter: Ø 25 mm Feedback Internal thread: G3/4" Recommended internal hose diameter: Ø 25 mm
Compressed air	Quick release coupling, nominal width NW 7.2 Input pressure: 6 bar Operating pressure: 4 bar
Volume of the vacuum storage tank	60 l
Volume of the liquid storage tank	30 l
Liquid feedback	3.5 litres/min
Electrical connections	Multi-pole plug for 400V supply Multi-pole plug for 24V supply
Compatible vacuum pumps	EVE-OG-63; EVE-OG-100 Only oil-lubricated vacuum pumps with gas ballast valves are recommended Additional pumps on request
Approximate weight	265 kg

3.2 Pneumatic Connection Diagram



4 Delivery, Packaging and Transport

4.1 Delivery

4.1.1 Scope of delivery

The precise scope of delivery is provided in the order confirmation.

The separately supplied oil must be added to the vacuum generator, item [2], in accordance with the accompanying operating instructions prior to commissioning the Vacuum-Operation-Center (VOC).

The weight and dimensions are listed in the delivery notes. This information must be complied with in order to ensure that appropriate means of transport and fastenings are used.



The operating instructions are part of the Vacuum-Operation-Center and must be included as part of any change of location.

4.1.2 Check integrity

Check that everything has been included in the shipment based on the enclosed delivery notes.

4.1.3 Report damage

Damage as a result of defective packaging or transport must be reported to the freight forwarder and J. Schmalz GmbH immediately after receipt of the shipment.

4.2 Packaging

The type of packaging depends on the transport route, the product dimensions and the weight of the shipment.

Carefully unwrap the packaging, remove foils, cartons, tensioning straps and packaging timber.



CAUTION

Knives or other sharp objects may damage hoses or cables

Danger of malfunction and leakage

► Take care when removing the packaging.



The packaging material must be disposed of in accordance with local laws and directives. Marked transport aids and locks must be removed.

4.2.1 Inland (land transport)

The VOC-AD-S-63/100 is shipped on a pallet with max. dimensions of 3000 x 1200 x 1000 mm.

The oil is drained from the pump and is enclosed with the shipment, packaged in a carton.

4.2.2 Abroad (overseas or air freight)

As described above, however in transport crates (wood) in sizes

approx. 2500 x 1100 x 1250 mm for the VOC-AD-S-63/100 instead of a pallet.

4.3 Transport

4.3.1 Shipment

The VOC is shipped horizontally, lashed to a pallet, ex-factory.

The oil is drained from the pump and is enclosed with the shipment, packaged in a carton.

When forwarding (horizontal) or returning the unit, drain the oil from the pump and reattach the transport locks.

4.3.2 Assembly and removal of the transport locks

The unit may only be assembled by qualified experts.

The VOC is equipped with transport lugs, item [15], for transport and assembly.

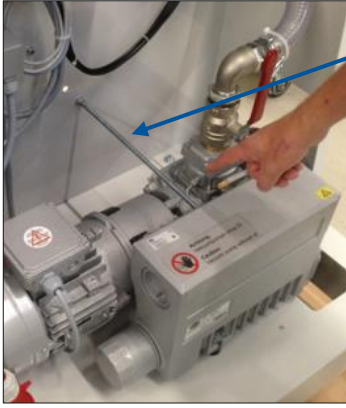
Lifting equipment must be attached to these lugs and the load must always be spread across all lugs.

 WARNING	
  	Suspended load, fall of the machine Risk of serious injury ▶ Take advantage of the lifting rings for installing suitable hoist for the transport. ▶ Never, go, stand or work under suspended loads. ▶ Use suitable means of transport. ▶ Safety clothing.

Set the unit down carefully and steadily. Jerky movements must be avoided.

Once the unit has been set down on the floor, the adjustable feet must be adjusted until the unit is level.

The VOC is delivered with a transport lock in the area of the vacuum generator, item [2], (cf. Fig. Transport lock). This transport lock must be removed from the vacuum generator, item [2], once the unit has been assembled and prior to commissioning.



Remove transport locks prior to commissioning the vacuum generator!

- Remove the threaded rod.

Fig.: Transport lock

5 Commissioning

The separately supplied oil must be added to the vacuum generator, item [2], in accordance with the accompanying operating instructions prior to commissioning the Vacuum-Operation-Center (VOC), pursuant to the display in the inspection glass.

ATTENTION	
	Operation of Vacuum-Operation-Centers without sufficient oil in the vacuum pump Damage to the Vacuum-Operation-Center ► Before commissioning and operation of the Vacuum-Operation-Center has the Oil level of the vacuum pump on the designated oil sight glass (9) to be controlled.

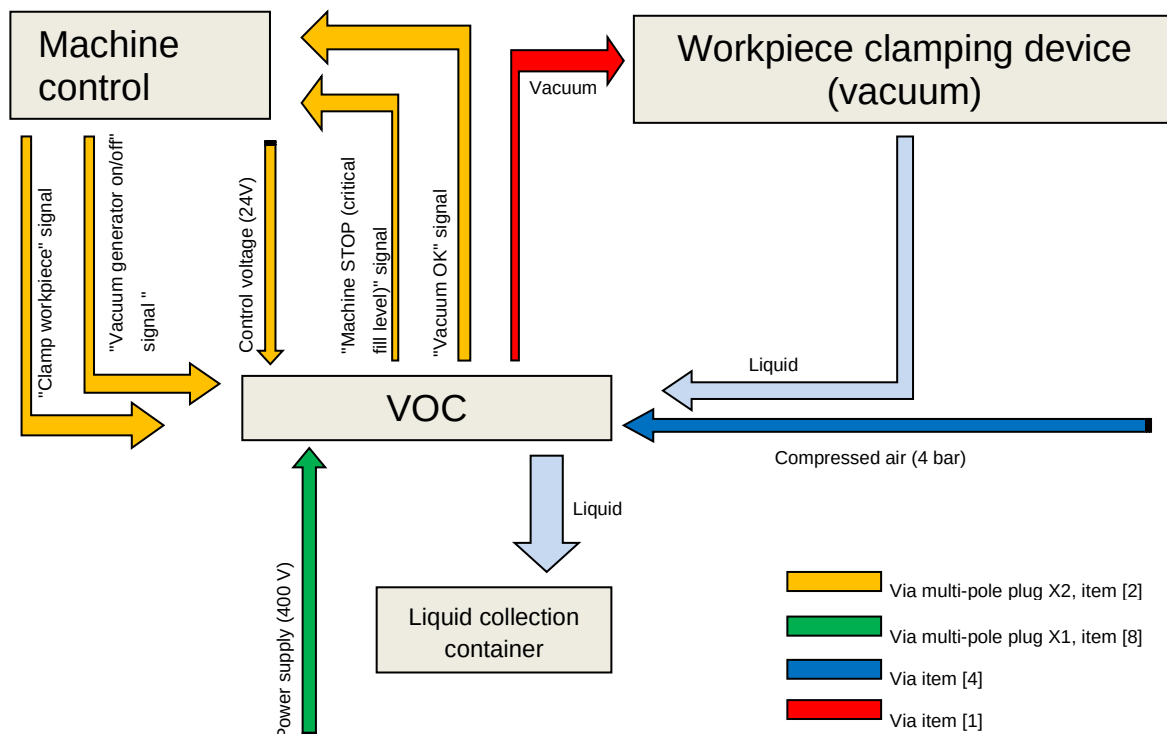
The VOC must be visually inspected for any damage prior to commissioning.

Please ensure that the VOC is positioned on a solid and level surface.

The VOC must be connected to the overall machine prior to commissioning. A description of the relevant designs and their integration into the system can be found in the following chapters.

5.1 Installation

Connect the unit in line with the following connection diagram:



5.1.1 Electrical connection

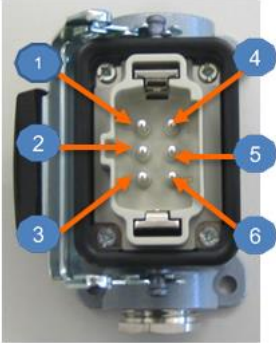
Power cables and other energy cables must be installed so that they do not present tripping hazards and they must be protected against mechanical damage. We recommend installing the cables in a closed conduit.

Two coded pins are inserted into each plug to ensure that the two multi-pole plugs (X1, X2) are not interchanged. These coded pins must be taken into account by the manufacturer of the overall machine when commissioning the VOC.

 DANGER	
 	Electric shock from contacting live parts Risk of death due to electric shock ▶ Work on the electrical installation may only be performed by an electrician. ▶ Disconnect the electrical energy supply prior to assembly, adjustment and maintenance activities and secure against restart. ▶ Ensure that the power has been disconnected, earthed and short-circuited. ▶ Do not connect or disconnect a live mains plug.

Electrical output data can be found on the pump motor's name plate item [2].

The connection is established via a coded multi-pole plug (X1) with 5 poles (400V, 16A), item [8].

Pin	Function	Representation of the pin assignment	Diagrammatic representation of the coding
PIN 1	L1		
PIN 2	L2		
PIN 3	L3		
PIN 4	N		
PIN 5	Not assigned		
PIN 6	Not assigned		
PE	PE		

For the VOC-AD-S-63 and -100, we recommend to pre-hedge a miniature fuse with 1,000 mA in the execution carries.

Check the direction of rotation of the vacuum pump

1. Switch on the Vacuum-Operation-Center from the overall machine control unit.
2. Switch on the vacuum generator from the overall machine control unit.
3. Observe the direction of rotation of the motor's fan blade, item [2]

Fan blade



IMPORTANT: The prescribed direction of rotation is displayed by an arrow on the motor housing.

4. The unit is shipped with a "clockwise field of rotation"

IMPORTANT: Immediately shut down the VOC if it is rotating in the wrong direction and reverse the polarity of the connection at the multi-pole plug (X1), item [8].

5. Check the direction of rotation.

ATTENTION**Operate the vacuum pump in the wrong direction of rotation**

Vacuum pump damaged

- Check the direction of rotation and reverse the polarity of the electrical supply lines if necessary.

5.1.2 Connecting the signal line to the overall machine

The signal line is connected via the coded multi-pole plug X2.

Pin	Function	Representation of the pin assignment	Diagrammatic representation of the coding
PIN 1	Vacuum pump ON/OFF		
PIN 2	Vacuum OK		
PIN 3	Machine stop		
PIN 4	Clamp/Release workpiece		
PIN 5	24V +		
PIN 6	24V -		
PE	PE		

The circuit diagram for the Vacuum-Operation-Center is located in the electrical cabinet.

Now integrate the signal into the machine control unit so that the workpiece can only be machined once the relevant signal has been received from the VOC and so that the current machining process is interrupted when the signal is no longer transmitted.

5.1.3 Connecting the vacuum and the liquid feedback

The Vacuum-Operation-Center must be connected to the clamping device via the vacuum connection, item [1], using an appropriate vacuum hose (cf. Chapter 3 "Technical Data").

The feedback of the liquid that has been sucked in via connection item [6] must also be connected to a suitable container with a vacuum hose (cf. Chapter 3 "Technical Data").

With the return of the coolant in the BAZ and the collecting an appropriate slope for liquid level of BAZ must be guaranteed..

5.1.4 Connecting the compressed air supply

The Vacuum-Operation-Center must be connected to the compressed air supply via an appropriate compressed air connection, item [4], (cf. Chapter 3 "Technical Data").

The compressed air must be filtered, dry and free of oil.

An input pressure of 6 bar is required to operate the VOC. This is restricted to the required operating pressure of 4 bar by the preset pressure regulator, item [9]. The VOC may only be operated at 4 bar.

IMPORTANT: The function of the valves, the seal and the targeted storage and feedback of the liquid is no longer ensured at an operating pressure of less than 4 bar.

IMPORTANT: An operating pressure above 4 bar results in the considerably higher wear of some components. The function of the valves, the seal and the targeted storage and feedback of the liquid is then no longer ensured.

5.2 Test run

When first commissioning the VOC, it is recommended that a test run without a workpiece is first performed for the entire system. Liquid is sucked in to check that the system sensors and actuators are functioning and that the signals are received and correctly interpreted by the overall machine control unit (BAZ).

Step	Action	Impact
S 1	Connect the VOC to the overall machine via the two multi-pole plugs	
S 2	Connect the compressed air, supply hose and feedback hose to the VOC	- The pressure regulator, item [9], must display 4 bar - The VOC is now ready
S 3	Switch on the VOC from the overall machine	- Valves 1V1 and 2V1 close - Vacuum switch 1Z2, item [11], displays the value -0.00
S 4	Start the vacuum generator from the overall machine	- Storage tank is evacuated - The vacuum value is shown by the display, item [14]
S 5	Evacuate the clamping device, incl. the liquid part (no workpiece and with slight leakage) To test the function of the fill level limit switches	- Valve 1V1 closes and valve 1V2 opens - Machining release from -600 mbar - Vacuum value for machining release is displayed by vacuum switch 1Z2, item [11] - Storage tank fills with liquid
S 6	The liquid feedback starts automatically	- Liquid is fed back in a targeted manner
S 7	Critical fill level reached	- Liquid level rises despite feedback - Machining stops in the overall machine and the liquid supply ceases - The vacuum generator continues to run
S 8	Liquid feedback	- Liquid continues to be fed back until liquid sensor 2Z1, item [12], is automatically reset
S 9	Reset liquid sensor 2Z1	- Release machine once again (el. signal)
S 10	Liquid is sucked in	- Storage tank fills
S 11	EMERGENCY STOP	- The EMERGENCY STOP is defined by the machine manufacturer

6 Operation

Before starting make sure that you are familiar with:

- The specific features of the VOC,
- The machine equipment,
- The operation of the VOC,
- The immediate environment in which the VOC and the machine or system, with which the VOC is used, is located,
- The machine's safety installations and
- The measures in case of emergency.

The following activities must be performed before starting:

- Check and ensure that all safety installations have been attached and are functioning.
- Visually inspect the VOC for damage and immediately remove any identified defects or report these to the supervisors (the VOC and the machine may only be operated in a flawless condition).
- Check and ensure that only authorised individuals are located in the machine's or the system's area of operation and that no other individuals are placed at risk by switching on the machine.

An operating vacuum of at least -600 mbar, which is monitored by vacuum switch [1Z2], item [11], must be reached to machine workpieces on the overall machine.

The workpiece being machined on the machine table or in the system may shift or even be released in the following circumstances:

- Power failure,
- Critical fill level exceeded,
- Pump failure and
- Line break.

 CAUTION	
	Danger from unexpected malfunctions Personal injury and/or property damage ▶ Restart the workpiece clamping process from the very beginning after a failure or disruption of the vacuum supply system. ▶ Restore the vacuum supply and ▶ Check the system for existing software errors (machine control switching functions)

The continuous operation of the VOC will heat the vacuum pump, item [2], and the surfaces can reach temperatures of over 70°C. Avoid contact with these surfaces.

Allow the vacuum pump to cool before performing maintenance work.

 CAUTION	
	Risk of burns from contact with hot surfaces Burns to the skin ▶ Do not touch hot surfaces ▶ Only perform maintenance work after allowing for a 1h cooling period

7 Troubleshooting

Fault	Possible cause	Remedy
Vacuum generator not working	- Vacuum generator defective - VOC not connected - Machine safety switch inactive	- Contact the service department - Connect the VOC to the mains - Check the status of the machine safety switch - Contact the service department
No liquid separation	- Pinch valves inactive - No compressed air supply to the solenoid valves - No vacuum supply to the solenoid valves - Solenoid valves inactive - Fill level sensor 2Z1 inactive	- Check solenoid valves using the manual override - Check the connection lines/contacts - Check the supply lines - Switch on the vacuum generator - Check the function of the fill level sensors using the LED display - Contact the service department
Insufficient vacuum in the storage tank	- Vacuum generator defective - Pinch valves inactive - Prefilter is clogged - Leakage in the VOC	- Contact the service department - Check the solenoid valves - Check the connecting lines - Clean or replace the prefilter - Check for leakage and remove
Insufficient/no vacuum for the clamping device	- The vacuum generator is not working - Vacuum not released - No vacuum exists in the storage tank - Leakage in the clamping device - Connection hose between the VOC and the clamping device are faulty or not installed - Gasket in the clamping device worn or not in place	- Connect the VOC to the mains - Switch on the vacuum generator - Release vacuum "Clamp workpiece" for the clamping device - Check VOC for seal tightness - Check clamping device for leakage. - Check clamping device for integrity (of seal) - Check seal for wear and replace if required - Check the connecting hose - Check seal for wear and replace if required

	- Prefilter is clogged - Pinch valves inactive	- Clean or replace prefilter - Check the connection lines - Check the compressed air and vacuum
Escape of liquid from the VOC	- Leakage in the storage tank	- Check for leakage and remove
Foam formation in the vacuum circuit or in the entire system	- Composition of the liquid is incorrect	- Check the liquid and replace if necessary
Liquid in the pump oil	- Water separator full - Liquid entering the vacuum generator was not separated - Gas ballast valves on the vacuum pump defective	- Drain and clean the water separator - Clean the oil (see the operating instructions for the rel. vacuum generator) - Replace the gas ballast valve



DANGER



Electric shock from contacting live parts

Risk of death due to electric shock

- ▶ Work on the electrical installation may only be performed by an electrician.
- ▶ Disconnect the electrical energy supply prior to assembly, adjustment and maintenance activities and secure against restart.
- ▶ Ensure that the power has been disconnected, earthed and short-circuited.
- ▶ Do not connect or disconnect a live mains plug.

8 Maintenance

8.1 General safety information



Please read Chapter 1.6 (Requirements and information relating to the installation, maintenance and operating personnel) before starting any maintenance work.



WARNING



Accident risk when maintenance is performed on the system by unqualified personnel

Consequences include serious personnel injury and property damage

- ▶ Only qualified personnel that have read and understood the operating instructions may perform maintenance work on the system.



DANGER



Electric shock from contacting live parts

Risk of death due to electric shock

- ▶ Work on the electrical installation may only be performed by an electrician.
- ▶ Disconnect the electrical energy supply prior to assembly, adjustment and maintenance activities and secure against restart.
- ▶ Ensure that the power has been disconnected, earthed and short-circuited.
- ▶ Do not connect or disconnect a live mains plug.



CAUTION



Risk of burns from contact with hot surfaces

Burns to the skin

- ▶ Do not touch hot surfaces
- ▶ Only perform maintenance work after allowing for a 1h cooling period

8.2 Pump oil maintenance

ATTENTION

No or incorrect maintenance of the pump oil

Destruction of the vacuum pump in case of loss of lubricity

- Compliance with maintenance regulations

The oil must be serviced at regular intervals in order to separate the oil in the vacuum generator from condensed liquids:

1. Set the vacuum generator in operation
2. Separate the vacuum generator, item [5], (figure in Chap. 8.3) from the vacuum circuit by closing the manual valve, item [6] (figure in Chap. 8.3).

Leave the vacuum generator in operation. This allows condensed liquids to be separated from the oil and transported out of the pump with the help of the gas ballast valves.

This condition should continue until the oil is cleaned, but at least 20 minutes.

This process must be repeated at regular intervals (cf. Chapter 8.3 Service Table).

Daily check the oil level when the pump is switched off. The oil level should not fall below sight glass center.

To ensure the pump function, an oil change by time intervals is necessary:

- First oil change after 100 hours of operation,
- and then every 500 to 1000 hours,
- at least every six months,
- or at too high fluid intake.



WARNING



Oil gets into the environment

Health hazards and environmental damage

- Proper disposal of used oil in accordance with the country-specific regulations.

In addition, please follow the maintenance information in the operating instructions TDS 30.30.01.00003 of the vacuum generator:

8.3 Service table



All maintenance activities refer to a 1 shift operation (8h). In the case of a 2 shift operation please adjust the service table accordingly!

	Interval		
	Weekly (A)	Monthly (B)	Quarterly (C)
Check the level of the droplet separator, item [1], and drain if necessary (A)	X		
Clean the prefilter, item [2], (B)		X	
Clean the inspection glass for the droplet separator, item [1], the fine filter, item [3], and the evacuation inspection glass, item [4], (B)		X	
Check the fine filter, item [3], and replace if necc. (C)			X
Oil change for the vacuum generator, item [5]	Cf chapter 8.2 "Pump oil maintenance"		
Overall system test run (C) (cf. Chapter 6.4 Test run)			X

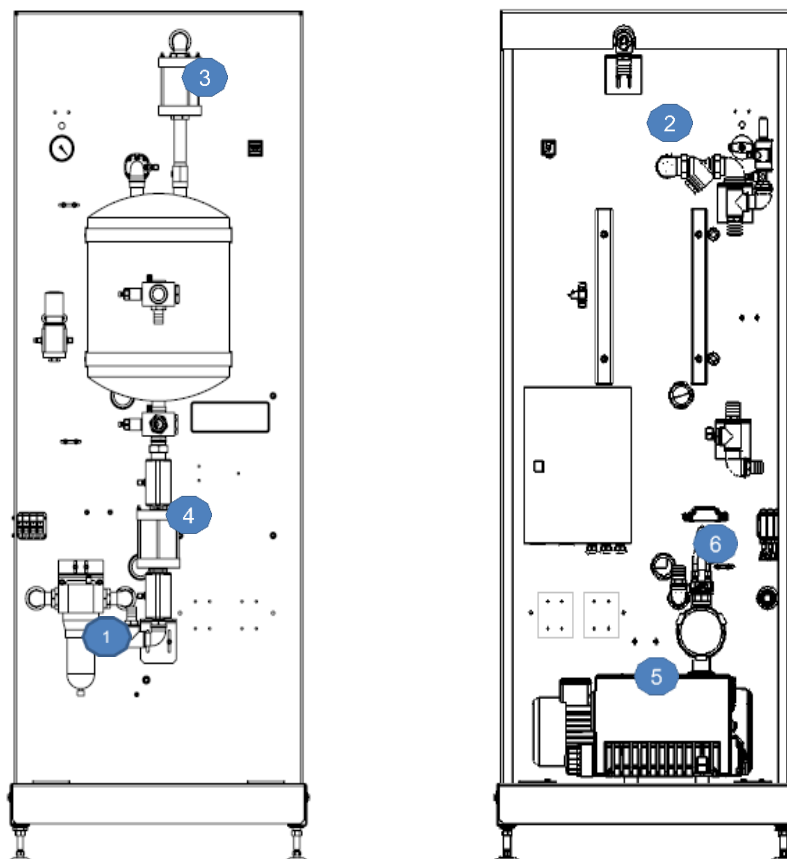


Fig.: VOC-AD-S-63/100



All maintenance activities must be signed and documented by qualified personnel.

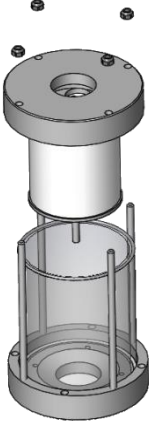
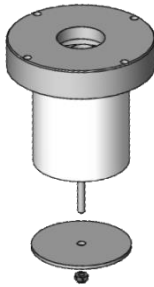
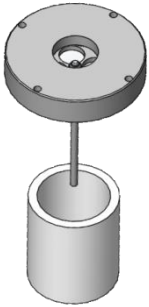
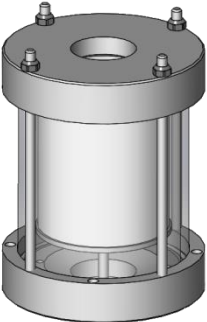
Year 20__

CW	Activity	Signature	Date	CW	Activity	Signature	Date
1	A			27	A		
2	A			28	A,B		
3	A			29	A		
4	A,B			30	A		
5	A			31	A		
6	A			32	A,B		
7	A			33	A		
8	A,B			34	A		
9	A			35	A		
10	A			36	A,B,C		
11	A			37	A		
12	A,B,C			38	A		
13	A			39	A		
14	A			40	A,B		
15	A			41	A		
16	A,B			42	A		
17	A			43	A		
18	A			44	A,B		
19	A			45	A		
20	A,B			46	A		
21	A			47	A		
22	A			48	A,B,C		
23	A			49	A		
24	A,B,C			50	A		
25	A			51	A		
26	A			52	A,B,C		

8.4 Cleaning and replacing the fine filter

Before starting this maintenance work, ensure that the VOC has been switched off.

The following table describes the progressive cleaning and replacement of the fine filter, item [3], in the figure in Chap. 8.3.

Step 1	Release the vacuum hose on top of the lid from the hose sleeve. Remove the four SW10 hexagonal nuts using a hexagon wrench. Remove the lid by pulling it upwards. The filter cartridge is also mounted to the lid.	
Step 2	Remove the SW10 hexagonal nuts underneath the retaining lid of the filter cartridge with a SW10 wrench and remove the lid.	
Step 3	Remove the filter cartridge by pulling it downwards and check for cleanliness. Clean or replace with a new filter cartridge if necessary.	
Step 4	Assemble the entire filter in reverse order.	

9 Spare and Wear Parts

9.1 Warranty

For this system, we give warranty in accordance with our General Terms and Conditions. The same applies to spare parts, provided they are delivered by us original parts.

For damages caused by the use of non-original spare parts or non-original accessories, we disclaim any responsibility on our part.

The exclusive use of original spare and wear parts is a prerequisite for the proper functioning of the Vacuum-Operation-Center and for warranty.

Excluded from the guarantee are all wear parts.

ATTENTION

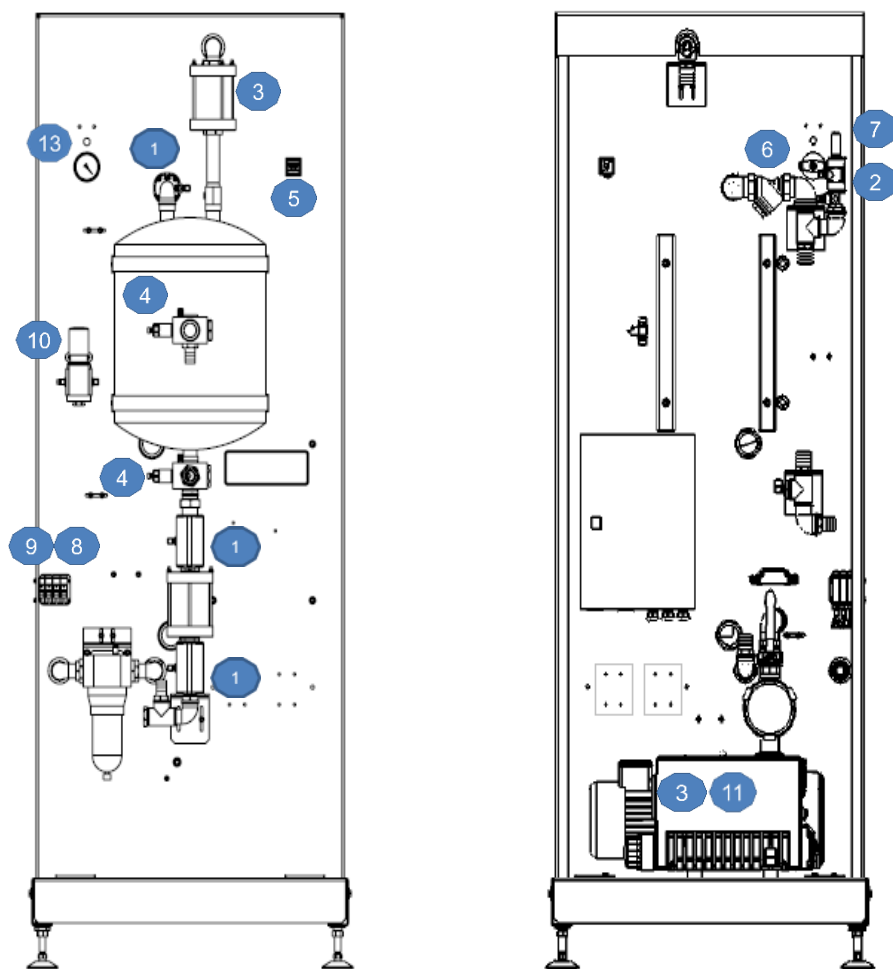
Use of non-original spare and wear parts

May result in malfunctions or property damage.

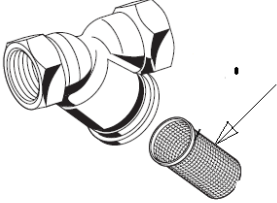
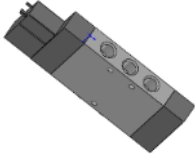
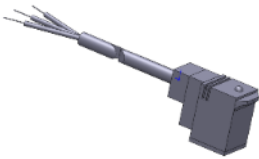
- ▶ Only use original parts provided by J. Schmalz as otherwise the warranty expires.

The following figures and tables display possible spare and wear parts.

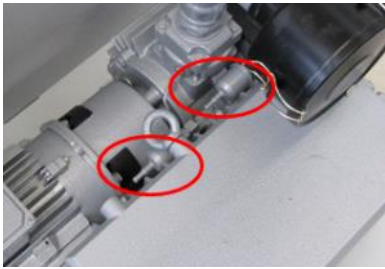
9.2 Spare and Wear Parts for VOC-AD-S-63/100



Item Chap. 9	Article	Description	Schmalz Art. No.	Type E=Spare part V=Wear part
1		Pinch valve QV G1"-IG	10.05.06.00086	V
2		Pinch valve QV G3/8-IG	10.05.06.00085	V
3		Filter	10.07.01.00310	V

4		Fill level limit switch NAEH-SCHA 24V-DC	21.01.09.00060	E
5		Vacuum switch VS-V-W-D-K-2C-PNP	10.06.02.00517	E
6		Replacement sieve ES 30-1 1/4A	10.01.27.00371	V
7		Silencer SD G3/8-AG 56 SEG	10.02.01.00905	V
8		Solenoid valve EMV 8 24V-DC 5/2 NO/NC 1.5	10.01.22.00158	E
9		Connection line ASK C 5000 PVC W-LED	21.04.06.00084	E
10		Pressure regulator Pressure range: 0.5 - 10.0 bar	10.07.11.00028	E

SPARE AND WEAR PARTS

11		Gas ballast valve	10.07.12.00072	V
12	Pump oil for EVE-OG Volume: 1 l	OEL EVE-OG 1L	27.02.01.00055	V
13		Vacuum-Manometer VAM 67 V H-SE	10.07.02.00016	E

10 Accompanying Documents

10.1 Vacuum Switch Operating Instructions

Operating instructions TDS 30.30.01.00721 apply for Vacuum-Operation-Centers in sizes VOC-AD-S-63/100.

10.2 Vacuum Generator Operating Instructions

Operating instructions TDS 30.30.01.00003 apply for Vacuum-Operation-Centers in sizes VOC-AD-S-63/100.

10.3 Declaration of Conformity

See next page

DE EG-Konformitätserklärung
EN EC-Declaration of Conformity
FR CE-Déclaration de conformité
ES Certificado de conformidad CE
IT Dichiarazione di conformità CE
NL CE Conformiteitsverklaring



Hersteller / Manufacturer / Fabricant / Fabricante / Produttore / Fabrikant

J. Schmalz GmbH, Aacher-Str. 29, D - 72293 Glatten

Produktbezeichnung / Product name / Designation du produit /
Denominación del producto / Denominazione del prodotto / Beschrijving van de machine

Produkt der Serie / Product, Series / Produit de la série / Producto, serie /
Prodotta, linea / product in de serie

Vacuum Operation Center Autark/ Vacuum control unit autarkic /
Vide Operation Centre autarchique/ Vacio Operation Center autárquico /
Vuoto Operation Centro autarchico/ Vacuum operation Center
autarkisch

Erfüllte einschlägige EG-Richtlinien / Applicable EC directives met / Directives CE applicables respectées /
Directivas vigentes de la CE cumplidas / Direttive CE applicate ed osservate / Nagekomen betreffende EG-richtlijnen

2006/42/EG Maschinenrichtlinie / Machinery Directive / Directive sur les machines /
Directiva para máquinas / Direttiva macchine / Machinerichtlijn
2004/108/EG Elektromagnetische Verträglichkeit / Electromagnetic Compatibility / Compatibilité électromagnétique /
Compatibilidad electromagnética / Compatibilità elettromagnetica / Elektromagnetische compatibiliteit
2006/95/EG Niederspannungsrichtlinie / Low Voltage Directive / Directive basse tension / Directiva de baja tensión / Direttiva sulla bassa tensione /
Laagspanningsrichtlijn
97/23/EG Druckgeräterichtlinie / Pressure Equipment Directive / Directive sur les équipements sous pression / Directiva de equipos a presión / Direttive per gli
apparecchi a pressione / Richtlijn drukapparatuur

Angewendete harmonisierte Normen / Harmonised standards applied / Normes d'harmonisation appliquées /
Normas armonizadas aplicadas / Norme armonizzate adottate / Toegepaste geharmoniseerde normen

EN ISO 12100 Sicherheit von Maschinen - Allgemeine Gestaltungsgrundsätze - Risikoabschätzung und Risikominderung / Safety of Machinery - General principles for
design - Risk assessment and risk reduction / Sécurité des machines - Principes généraux de conception - Appréciation du risque et réduction du
risque / Seguridad de máquinas - Principios generales de diseño - Evaluación del riesgo y reducción del riesgo / Sicurezza delle macchine - Principi
generali di progettazione - Valutazione del rischio e riduzione del rischio / Veiligheid van machines - Algemene beginselen voor ontwerp -
Risicoabschätzung en de risico-reductie
EN 61000-6-3 Elektromagnetische Verträglichkeit - Störaussendung / Electromagnetic Compatibility - Emission /
Compatibilité électromagnétique - Norme sur l'émission / Compatibilidad electromagnética - Emisión de interferencias /
Compatibilità elettromagnetica - Norma generale sull'emissione / Elektromagnetische compatibiliteit - emissie
EN 61000-6-2 Elektromagnetische Verträglichkeit - Störfestigkeit / Electromagnetic Compatibility - Immunity /
Compatibilité électromagnétique - Immunité / Compatibilidad electromagnética - Resistencia a interferencias /
Compatibilità elettromagnetica - Immunità / Elektromagnetische compatibiliteit - immunitet
EN 60204-1 Sicherheit von Maschinen - Elektrische Ausrüstung von Maschinen / Safety of Machinery - Electrical equipment of machines / Sécurité des machines
- Equipement électrique des machines / Seguridad de máquinas - Equipamiento eléctrico de máquinas / Sicurezza delle macchine -
Equipaggiamento elettrico delle macchine / Veiligheid van machines - elektrische uitrusting van machines
EN ISO 13857 Sicherheit von Maschinen - Sicherheitsabstände gegen das Erreichen von Gefährdungsbereichen mit den oberen und unteren Gliedmaßen / Safety
of Machinery - Safety distances to prevent hazard zones being reached by upper and lower limbs / Sécurité des machines - Distances de sécurité
empêchant les membres supérieurs et inférieurs d'atteindre les zones dangereuses / Seguridad de máquinas - Distancias de seguridad contra el
contacto de los miembros superiores e inferiores con zonas de peligro / Sicurezza delle macchine - Distanze di sicurezza per prevenire che le parti
superiori ed inferiori raggiungano le aree di pericolo / Veiligheid van machines - veiligheidsafstanden ter voorkoming van het bereiken van
gevaarlijke zones met de bovenste en onderste ledematen
EN ISO 2151 Akustik- Kompressoren und Vakuumpumpen, Bestimmung der Geräuschemission / Acoustics - Noise test code for compressors and vacuum pumps
/ Acoustique - Code d'essai acoustique pour les compresseurs et les pompes à vide / Acústica - Compresores y bombas de vacío, medición del
ruido / Acustica di compressori e pompe per vuoto, determinazione dell'emissione di rumori / Akoestiek - Compressoren en vacuumpompen -
Bepaling van geluidsemissie
EN ISO 13849-1 Sicherheit von Maschinen - Sicherheitsbezogene Teile von Steuerungen / Safety of machinery - Safety-related parts of control systems / Sécurité
des machines - Parties des systèmes de commande relatives à la sécurité / Seguridad de máquinas - Componentes seguros en sistemas de control
/ Sicurezza delle macchine - Parti/Componenti di sistemi di comando relativi alla sicurezza / Veiligheid van machines - Onderdelen van
besturingssystemen met een veiligheidsfunctie
EN 1012-1 Kompressoren und Vakuumpumpen - Sicherheitsanforderungen / Compressors and Vacuum Pumps - Safety requirements / Compresseurs et
pompes à vide - Prescriptions de sécurité / Compresores y bombas de vacío - Requisitos de seguridad / Compressori e pompe per vuoto - requisiti
di sicurezza / Compressoren en vacuumpompen - veiligheidsisen



Dokumentationsverantwortlicher / Person responsible for documentation / Responsable de la documentation /
Responsable de documentación / Responsabile della documentazione / Verantwoordelijk voor de documentatie

Glatten, 28.04.2014, K.D. Fanta
Klaus-Dieter Fanta / J. Schmalz GmbH, Aacher-Str. 29, D - 72293 Glatten

Unterschrift, Angaben zum Unterzeichner / Signature, details of signatory / Signature, indications sur le soussigné /
Firma y datos del firmante / Firma, dati concernenti il firmatario / Handtekening, omschrijving van de ondertekenaar

Glatten, 28.04.14, Ralf Stockburger
Ralf Stockburger
Leiter Geschäftsentwicklung Vakuumaufspannsysteme /
Head of Business Development Vakuum Clamping Systems

Schmalz Services



Contact worldwide

Our sales network with local sales representatives, international branches and trading partners guarantees quick and competent information and advice in over 50 countries worldwide.

 www.schmalz.com/vertriebsnetz



Documents online

Download catalogues, operating instructions and CAD data conveniently online and familiarise yourself with our products and services.

www.schmalz.com/dokumentationen



"Know how..." videos

The short videos provide a straightforward and clear explanation of the extensive functions of our products. Take a look, it's worth it!

 www.schmalz.com/gewusst-wie

Additional services, from consulting through to training can be found at

 www.schmalz.com/services